Amazon S3 – Bucket Policy

The policy JSON file contains a series of strings and values. The strings that are used will vary depending on what the policy is designed to do. Most policies start out with three strings:

- **Version:** Usually the date that the policy was created or modified.
- **ID:** A policy name.
- **Statement:** Designates that a block of policy settings is to follow. These policy settings are enclosed in brackets.

This portion of the policy looks something like this:

```
{
   "Version": "2016-09-21",
   "Id": "Posey Sample Policy",
   "Statement": [
     {
      The policy settings go here.
     }
   ]
}
```

As previously noted, the policy settings that are used can vary depending on what the policy is intended to do. There are five strings that are used in most cases. These five strings are:

- **Sid:** The SID basically determines what the policy does, and should not be confused with a Windows Security Identifier. For example, if the action that needs to be performed involves adding a user to an Access Control List (ACL), then the SID would be AddCannedAcl. Similarly, if the policy is designed to evaluate IP addresses, then the SID might be set to IPAllow. The previously mentioned Examples page lists a variety of SIDs that you can use.
- **Effect:** The Effect string controls what happens when the policy is applied. In most cases, the effect can be to either allow an action or to deny an action.
- **Principal:** The Principal string controls who the policy applies to. Setting the Principal string to * makes the string apply to everyone, but it is possible to specify individual AWS accounts.

- **Action:** The Action is what happens if the policy is applied. For example "s3:GetObject" allows the principal to read object data.
- **Resource:** The Resource is the bucket to which the statement applies. You cannot, however, simply enter a bucket name. The bucket name must be entered in a special format. If, for example, the statement were to apply to a bucket named poseydemo-test, then the resource would be expressed as "arn:aws:s3:::poseydemo-test/*".

Here is an example of a very simple bucket policy:

```
{
  "Version":"2016-09-22",
   "Id": "Posey Sample Policy",
  "Statement":[
    {
      "Sid":"AddPerm",
      "Effect":"Allow",
      "Principal": "*",
      "Action":["s3:GetObject"],
      "Resource":["arn:aws:s3:::poseydemo-test/*"]
    }
  ]
}
```

This simple policy (which was adapted from one of the examples shown in the AWS documentation) grants the s3:GetObject permission to public anonymous users, thereby allowing them to read the bucket contents. Such a policy might be useful if the bucket contained content for a public Web site.

As you can see, bucket policies allow for tighter control over bucket permissions. It is even possible to build conditional policies and multi-statement policies.

Reference Links:

https://docs.aws.amazon.com/AmazonS3/latest/user-guide/set-permissions.html

https://docs.aws.amazon.com/AmazonS3/latest/dev/example-policies-s3.html

https://awspolicygen.s3.amazonaws.com/policygen.html

https://docs.aws.amazon.com/AmazonS3/latest/user-guide/what-is-s3.html